

Sets and Sample Spaces

Jericho Lawson

MATH 241

Lecture #2

Reminders!

- Homework #0 is due on Friday, 9/4 at 11:59pm
 - ▶ Includes survey, setup of tech, and turning in quick problem
- Read through Chapter 1 of Introduction to Probability
- Office hours: none on Monday, Tuesday from 3-4:30pm in Rocky 106A

Learning Objectives

- **Familiarize** ourselves with the purpose of events and sample spaces
- **Visualize** subsets of the sample space when intersections, unions, and complements of event(s) are used
- **Introduce** techniques for proving theorems in general with the use of the proof of DeMorgan's laws

Sets

A set is a Many that allows itself to be thought of as a One.

-Georg Cantor

Sets

- A collection of objects
- Denoted as S
- $x \in S$ if it is an element of the set
 - ▶ How would the notation change if x was not an element of S ?

Sets

Set of all even numbers $\rightarrow \{0, 2, 4, 6, 8, \dots\}$

Set of places Jericho has lived in $\rightarrow \{\text{Phoenix, Riverside, } \dots\}$

All real numbers between 2 and 8 $\rightarrow [2, 8]$

Real coordinates within one unit of center $\rightarrow \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$

Set of all outcomes when rolling 2 dice $\rightarrow$?

Set of all weather events that could occur today $\rightarrow$?

Other Set-Related Items

- Empty set: $\{\}, \emptyset$
 - ▶ Denotes nothing existing inside
- Power set: $\mathcal{P}(S)$
 - ▶ Set of all subsets of S
- Subset: $A \subseteq B$
 - ▶ Every element of A must be an element of B
- Partition: $A_1 \cup A_2 \cup \dots \cup A_n = S$
 - ▶ Collection of disjoint subsets; union is entire set
- Cardinality: $|A|$
 - ▶ Size; number of elements in set

Sample Spaces

Definition

The sample space S of an experiment is the set of all possible outcomes of the experiment.

Examples

Birthdays of n people in a room $\rightarrow S$ has 365^n outcomes*

Outcomes of two dice rolled $\rightarrow S$ has 36 outcomes

Infection status of two people $\rightarrow S$ has 4 outcomes

Example of Two Dice Rolled

1st Dice	2nd Dice					
	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
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- $S = \{(1, 1), (1, 2), \dots, (6, 6)\}$
- $S = \{(i, j) : 1 \leq i \leq 6, 1 \leq j \leq 6\}$
- $|S| = 36$

Event

Definition

An event E is any subset of the sample space; that is, it's a set consisting of possible outcomes of the sample space. If an outcome in E has occurred, then we say the event E has occurred.

Examples

- Birthdays of n people in a room
 - ▶ E = at least two people that share the same birthday
- Outcomes of two dice rolled
 - ▶ E = sum is odd
- Infection status of two people
 - ▶ E = both people are infected

Example of Two Dice Rolled

- E = sum is odd
- F = sum is 5

Example of Two Dice Rolled

- E = sum is odd
- F = sum is 5

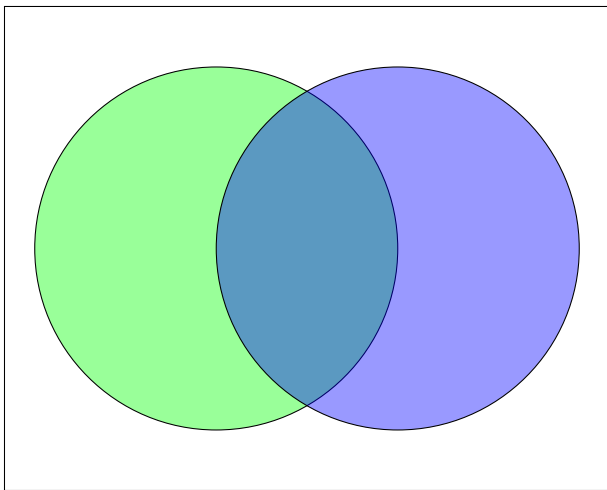
	2nd Dice					
1st Dice	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
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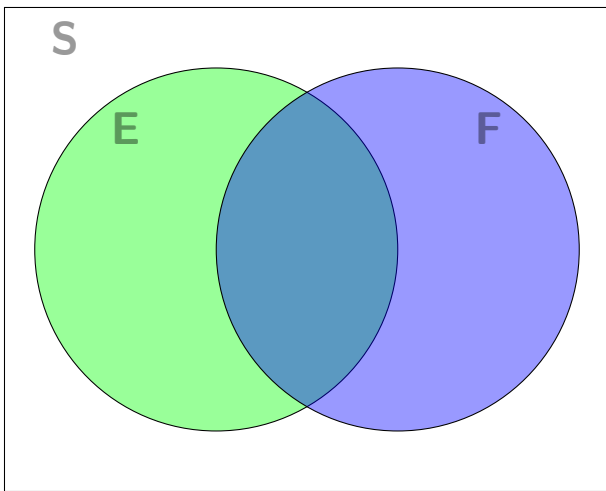
Example of Two Dice Rolled

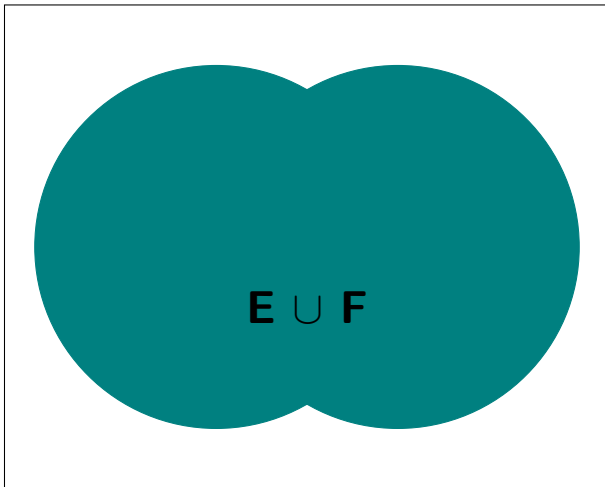
- E = sum is **odd**
- F = sum is **5**

	2nd Dice					
1st Dice	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
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Wait a minute?! Elements of the sample space exist in both events.
What does this mean??

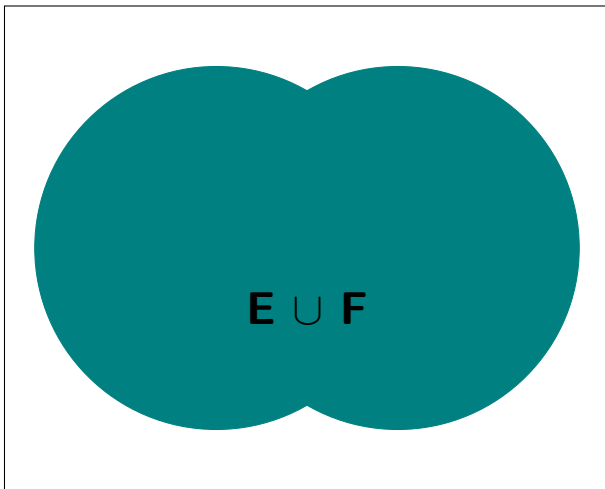






Definition

The union of two events E and F , which we write as $E \cup F$, is the event containing all outcomes in E or F , i.e. $E \cup F = \{x | x \in E \text{ or } x \in F\}$.



- Inclusive “or”
- An element in at least one event

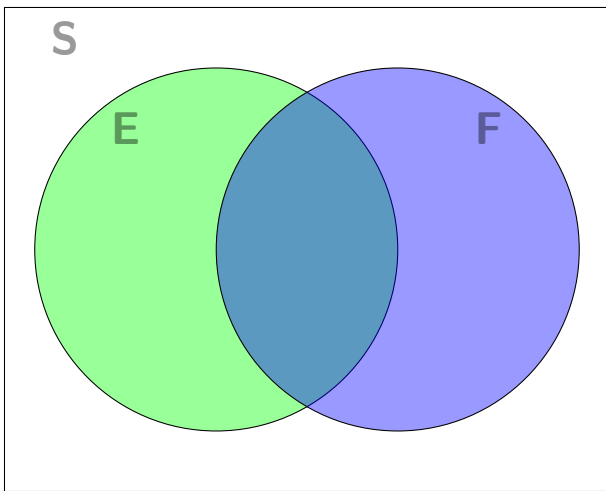
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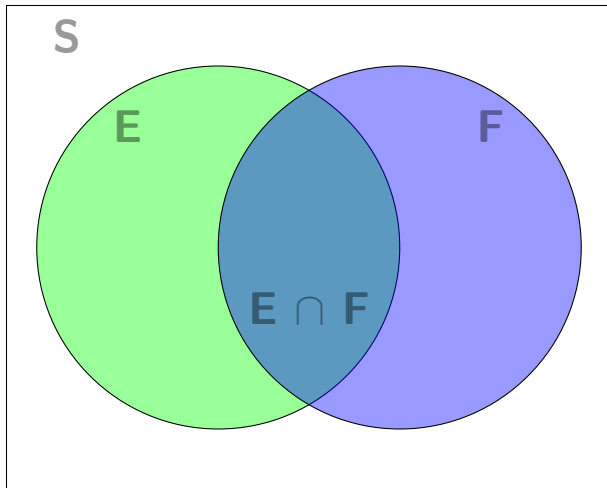
- E = sum is odd
- F = sum is 5
- $E \cup F$ = sum is **odd** or **5**

Example of Two Dice Rolled

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- F = sum is 5
- $E \cup F$ = sum is **odd** or **5**

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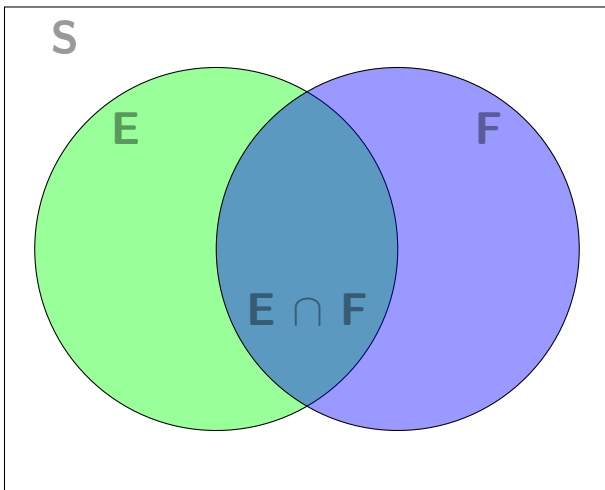




Definition

The intersection of two events E and F , which we write as $E \cap F$, is the event containing all outcomes in E and F , i.e.

$$E \cap F = \{x | x \in E \text{ and } x \in F\}.$$



- “and”
- An element in all events

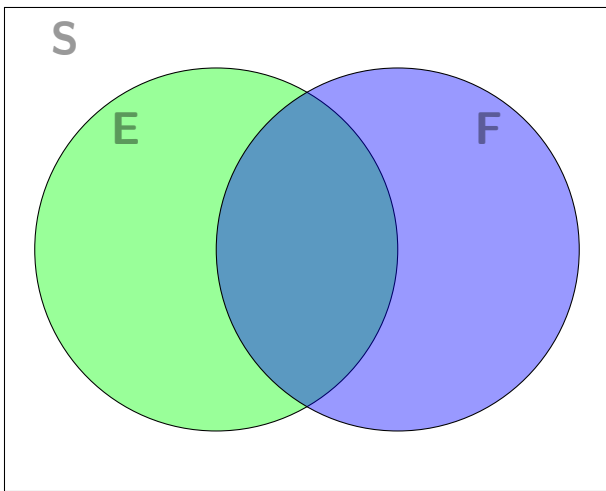
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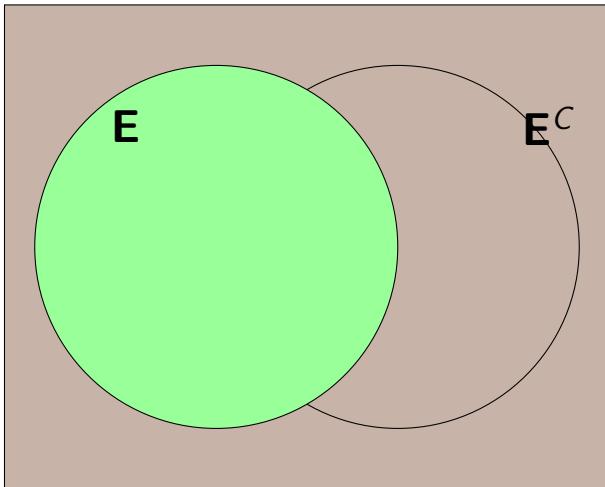
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- $E \cap F$ = sum is **odd and 5**

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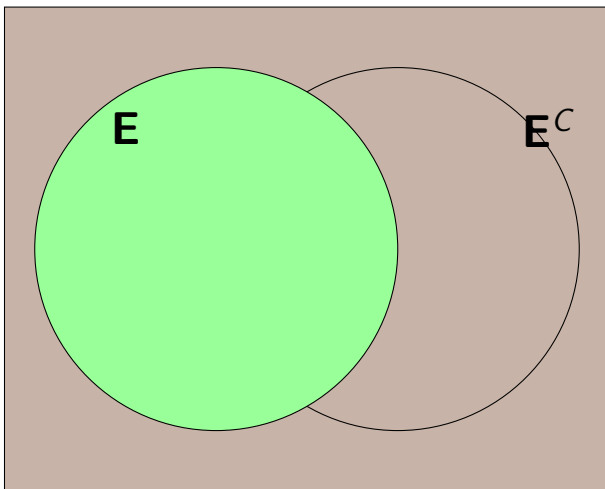
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Definition

The complement of E , which we write as E^C , is the event containing all outcomes in the sample space that are not in E , i.e. $E^C \cap F = \{x | x \notin E\}$.



- “not”
- $\setminus$ can be used to remove from an event (i.e. $E \setminus F$)
- Useful in situation where studying event E is more difficult

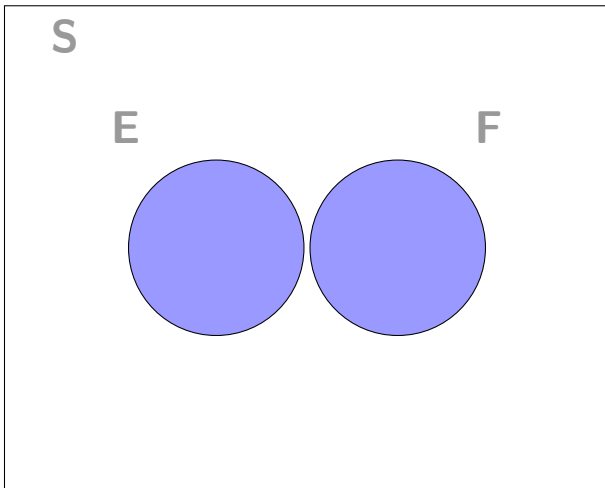
Example of Two Dice Rolled

- E = sum is odd
- F = sum is 5
- $E \notin S$ = sum is not **odd** and not **5**

Example of Two Dice Rolled

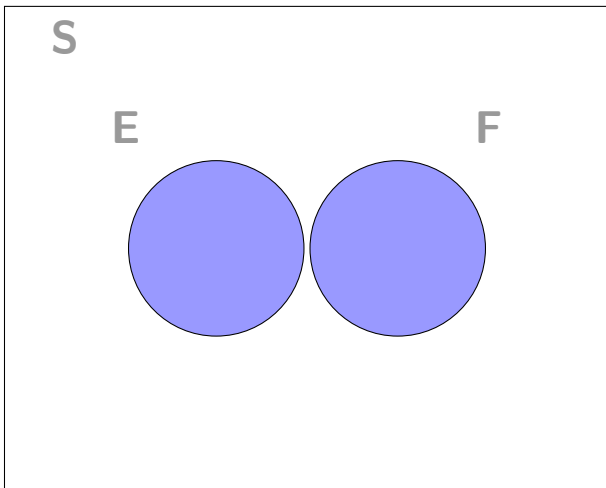
- E = sum is odd
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- $E \notin S$ = sum is not **odd** and not **5**

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Definition

Two events E and F are mutually exclusive (i.e. disjoint) if $E \cap F = \emptyset$.



- No overlap between two events

Example of Two Dice Rolled

- E = sum is odd
- F = sum is 4

Example of Two Dice Rolled

- E = sum is odd
- F = sum is 4

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Let's Combine Them!

- Naturally, we can combine notation together to represent certain spaces/subsets
 - ▶ e.g. $E \cup F^C$
 - ▶ e.g. $(E \cap F^C) \cup (E^C \cap F)$
 - ▶ e.g. S^C

Union and Intersection, Generalized

We can define the union and intersection for events $E_1, \dots, E_n$:

- Union: $\bigcup_{i=1}^n E_i$
- Intersection: $\bigcap_{i=1}^n E_i$

For countably infinite events, we can replace n with ∞ .

Set Operations

- Communative (flip order)
 - ▶ $E \cup F = F \cup E$
 - ▶ $E \cap F = F \cap E$
- Associative (groupings)
 - ▶ $E \cup (F \cup G) = (E \cup F) \cup G$
 - ▶ $E \cap (F \cap G) = (E \cap F) \cap G$
- Distributive (apply within groupings)
 - ▶ $E \cup (F \cap G) = (E \cup F) \cap (E \cup G)$
 - ▶ $E \cap (F \cup G) = (E \cap F) \cup (E \cap G)$

Proofs

- We will be writing proofs periodically throughout this course
 - ▶ Builds intuition about material
 - ▶ Formal mathematical language
- We will slowly go through the steps in the coming lectures

Set Equality

To prove that two sets A and B are equal to each other (i.e. $A = B$), we need to show containment in both directions, i.e.

$$A = B \iff A \subseteq B \text{ and } B \subseteq A$$

- $A \implies B$: if A , then B
- $A \iff B$: if and only if; if A , then B , and if B , then A

DeMorgan's Laws

For two events E_1 and E_2 ,

$$\textcircled{1} \quad (E_1 \cup E_2)^C = E_1^C \cap E_2^C$$

$$\textcircled{2} \quad (E_1 \cap E_2)^C = E_1^C \cup E_2^C$$

DeMorgan's Laws

For two events E_1 and E_2 , suppose E_1 is a sophomore in this classroom and E_2 is a math/stats major. Then,

① $(E_1 \cup E_2)^C = E_1^C \cap E_2^C$

- ▶ “Those that aren’t sophomore or math/stats majors” are the same as “those that aren’t sophomore and those that aren’t a math/stats major”.

② $(E_1 \cap E_2)^C = E_1^C \cup E_2^C$

- ▶ “Those that aren’t sophomore and math/stats majors” are the same as “those that aren’t sophomore or those that aren’t a math/stats major”.

To prove the first law:

$$(E_1 \cup E_2)^C = E_1^C \cap E_2^C$$

Proof of First Law

We need to show set containment in both directions:

- ① $(E_1 \cup E_2)^C \subseteq E_1^C \cap E_2^C$
- ② $E_1^C \cap E_2^C \subseteq (E_1 \cup E_2)^C$

$$(E_1 \cup E_2)^C = E_1^C \cap E_2^C$$

Proof of First Law (cont.)

We first show $(E_1 \cup E_2)^C \subseteq E_1^C \cap E_2^C$. Let x be an element in $(E_1 \cup E_2)^C$. Then,

$$\begin{aligned} x &\in (E_1 \cup E_2)^C \\ \implies x &\notin E_1 \cup E_2 && \text{by def. of complement} \\ \implies x &\notin E_1 \text{ and } x \notin E_2 && \text{by def. of union} \\ \implies x &\in E_1^C \text{ and } x \in E_2^C && \text{by def. of complement} \\ \implies x &\in E_1^C \cap E_2^C && \text{by def. of intersection} \end{aligned}$$

Since $x \in E_1^C \cap E_2^C$, then $(E_1 \cup E_2)^C \subseteq E_1^C \cap E_2^C$.

$$(E_1 \cup E_2)^C = E_1^C \cap E_2^C$$

Proof of First Law (cont.)

We now show $(E_1^C \cap E_2^C \subseteq E_1 \cup E_2)^C$. Let x be an element in $E_1^C \cap E_2^C$. Then,

$$\begin{aligned} x &\in E_1^C \cap E_2^C \\ \implies x &\in E_1^C \text{ and } x \in E_2^C && \text{def. of intersection} \\ \implies x &\notin E_1 \text{ and } x \notin E_2 && \text{by def. of complement} \\ \implies x &\notin E_1 \cup E_2 && \text{by def. of union} \\ \implies x &\in (E_1 \cup E_2)^C && \text{by def. of complement} \end{aligned}$$

Since $x \in (E_1 \cup E_2)^C$, then $E_1^C \cap E_2^C \subseteq (E_1 \cup E_2)^C$.

Since $(E_1 \cup E_2)^C \subseteq E_1^C \cap E_2^C$ and $E_1^C \cap E_2^C \subseteq (E_1 \cup E_2)^C$, then $(E_1 \cup E_2)^C = E_1^C \cap E_2^C$. $\square$

On Your Own Time

Prove DeMorgan's second law, using techniques seen in the proof of the first law.

DeMorgan's Laws, Generalized

DeMorgan's laws can be extended to a finite number of events n , as well as countably infinite number of events. For a finite number of events:

$$① \left(\bigcup_{i=1}^n E_i \right)^C = \bigcap_{i=1}^n E_i^C$$

$$② \left(\bigcap_{i=1}^n E_i \right)^C = \bigcup_{i=1}^n E_i^C$$

Naive Definition of Probability

Suppose all outcomes of an experiment are equally-likely, then the probability of an event A is given by:

$$P(A) = \frac{\text{number of outcomes in } A}{\text{number of outcomes in } S}$$

- Definition is helpful, but not widely applicable
 - ▶ Why?
- Next class: axiomatic definition of probability

See You Tuesday!

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